

EARLTON, ON, Canada

WMO: 716840

Lat: 47.6976N

Lon: 79.8535W

Elev: 244

StdP: 98.43

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-28.0	-35.2	0.1	-30.8	-31.9	0.2	-27.9	11.4	-8.2	10.5	-9.6	2.5	310	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.5	20.8	27.6	19.6	25.9	18.5	22.1	27.2	21.0	25.4	20.0	23.9	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.7	24.5	19.5	14.7	23.4	18.5	13.8	22.3	66.2	27.5	62.2	25.4	58.5	24.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.2	8.3	DB	-36.3	32.7	3.8	1.8	-39.1	33.9	-41.3	35.0	-43.4	35.9	-46.2	37.2	(4)
(5)			WB	-36.5	24.0	3.8	1.2	-39.2	24.9	-41.4	25.6	-43.5	26.3	-46.3	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-14.4	-13.1	-6.5	2.1	10.9	16.0	18.4	17.2	13.0	5.9	-1.6	-9.9	(6)	
	DBStd	12.89	8.15	6.77	6.85	5.37	4.88	3.99	3.26	3.37	4.55	4.71	5.75	7.34	(7)	
	HDD10.0	3345	758	648	513	242	49	3	0	0	20	147	348	618	(8)	
	HDD18.3	5613	1016	881	771	488	236	93	40	62	169	386	597	876	(9)	
	CDD10.0	876	0	0	1	4	76	182	259	224	110	19	1	0	(10)	
	CDD18.3	103	0	0	0	0	4	21	41	27	9	1	0	0	(11)	
	CDH23.3	1066	0	0	1	3	71	245	409	252	82	4	0	0	(12)	
	CDH26.7	266	0	0	0	0	16	64	108	62	15	1	0	0	(13)	
(14) Wind	WSAvg	3.9	4.0	4.1	4.1	4.4	4.2	3.6	3.3	3.3	3.6	4.0	4.2	4.0	(14)	
(15) Precipitation	PrecAvg	754	46	38	42	47	68	79	95	72	87	75	60	46	(15)	
	PrecMax	894	88	106	97	75	136	118	162	124	168	115	97	80	(16)	
	PrecMin	651	24	10	2	16	35	44	43	20	25	21	26	22	(17)	
	PrecStd	67	18	23	27	16	30	24	31	29	31	30	18	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.1	13.5	21.6	29.0	31.1	32.0	31.1	28.5	22.5	13.9	6.0	(19)	
		MCWB	3.0	2.4	10.3	13.6	18.9	21.4	22.4	21.8	21.0	16.6	10.8	4.4	(20)	
	2%	DB	1.7	1.4	7.4	16.9	25.0	28.5	29.4	28.4	25.7	19.2	10.5	3.1	(21)	
		MCWB	1.2	0.4	4.3	10.1	15.4	19.7	20.8	20.5	19.1	14.7	8.5	2.3	(22)	
	5%	DB	0.3	-0.5	5.1	13.5	22.6	26.2	27.5	26.1	23.5	16.4	8.4	1.5	(23)	
		MCWB	-0.3	-1.4	2.5	8.2	14.9	18.1	19.7	19.1	17.9	13.1	6.8	0.8	(24)	
	10%	DB	-1.8	-2.5	2.9	10.2	20.0	23.9	25.6	24.1	21.2	13.6	6.2	0.2	(25)	
		MCWB	-2.4	-3.3	1.0	6.1	13.6	17.1	18.8	18.6	17.3	11.3	4.9	-0.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.6	11.1	14.9	20.9	23.1	24.2	23.0	22.0	17.8	11.6	5.0	(27)	
		MCDB	3.6	3.6	12.8	20.7	25.4	29.0	29.8	28.5	26.4	22.0	13.4	5.5	(28)	
	2%	WB	1.3	0.5	4.7	11.0	18.0	21.2	22.2	21.5	20.5	15.4	8.8	2.4	(29)	
		MCDB	1.6	1.3	6.7	15.1	22.5	26.3	26.9	26.5	24.0	18.3	9.9	2.9	(30)	
	5%	WB	-0.3	-1.4	2.7	8.7	16.2	19.7	21.0	20.4	19.2	13.6	6.9	0.9	(31)	
		MCDB	0.3	-0.6	4.5	12.6	20.6	24.0	25.1	24.4	22.3	16.0	8.2	1.4	(32)	
	10%	WB	-2.4	-3.3	1.1	6.5	14.1	18.2	19.8	19.3	17.6	11.4	4.9	-0.2	(33)	
		MCDB	-1.7	-2.6	2.6	9.9	18.8	22.2	24.1	22.8	20.7	13.5	6.1	0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	11.4	10.4	12.5	11.9	11.8	11.3	11.2	8.5	7.1	8.5	(35)	
		MCDBR	8.0	10.5	12.0	16.5	16.2	15.2	14.5	14.4	14.3	13.2	10.0	7.2	(36)	
	5% WB	MCWBR	7.7	9.7	9.2	10.6	8.8	8.0	7.3	7.5	8.3	9.1	8.4	6.9	(37)	
		MCWBR	7.9	10.2	10.4	15.0	13.7	12.6	12.3	12.2	11.8	12.3	9.3	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.265	0.282	0.314	0.348	0.363	0.385	0.397	0.386	0.364	0.329	0.297	0.264	(40)	
		taud	2.283	2.225	2.244	2.344	2.397	2.355	2.350	2.373	2.417	2.502	2.481	2.362	(41)	
	Ebn at Noon		816	878	903	904	900	878	861	856	843	824	784	777	(42)	
		Edh at Noon	74	98	114	115	114	120	119	111	97	76	63	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.24	2.29	3.78	4.70	5.25	5.64	5.52	4.63	3.51	2.08	1.10	0.88	(44)	
	RadStd		0.19	0.34	0.40	0.41	0.43	0.38	0.45	0.31	0.33	0.27	0.12	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-0.73	-0.72	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page